

Unit - 1

Foundations of Design

UI vs UX Design- Core Stages of Design Thinking – Divergent and Convergent Thinking – Brain storming and Game Storming – Observational Empathy

What is user interface (UI)?

The user interface (UI) is the point of human-computer interaction and communication in a device. This can include display screens, keyboards, a mouse and the appearance of a desktop. It is also the way through which a user interacts with an application or a website.

The growing dependence of many businesses on web applications and mobile applications has led many companies to place increased priority on UI in an effort to improve the user's overall experience.

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Types of user interfaces

The various types of user interfaces include:

- graphical user interface (GUI)
- command line interface (CLI)
- menu-driven user interface
- touch user interface
- voice user interface (VUI)
- form-based user interface
- natural language user interface

Examples of user interfaces

Some examples of user interfaces include:

- computer mouse

- remote control
- virtual reality
- ATMs
- speedometer
- the old iPod click wheel

Websites such as Airbnb, Dropbox and Virgin America display strong user interface design. Sites like these have created pleasant, easily operable, user-centered designs (UCD) that focus on the user and their needs.

History of UI

In early computers, there was very little user interface except for a few buttons at an operator's console. Many of these early computers used punched cards, prepared using keypunch machines, as the primary method of input for computer programs and data. While punched cards have been essentially obsolete in computing since 2012, some voting machines still use a punched card system.

The user interface evolved with the introduction of the command line interface, which first appeared as a nearly blank display screen with a line for user input. Users relied on a keyboard and a set of commands to navigate exchanges of information with the computer. This command line interface led to one in which menus (lists of choices written in text) predominated.

Finally, the GUI arrived, originating mainly in Xerox's Palo Alto Research Center (PARC), adopted and enhanced by Apple and effectively standardized by Microsoft in its Windows operating systems.

Graphical UIs

Elements of a GUI include such things as windows, pull-down menus, buttons, scroll bars and icons. With the increasing use of multimedia as part of the GUI, sound, voice, motion video and virtual reality are increasingly becoming the GUI for many applications.

Mobile UIs

The emerging popularity of mobile applications has also affected UI, leading to something called mobile UI. Mobile UI is specifically concerned with creating usable, interactive interfaces on the smaller screens of smartphones and tablets and improving special features, like touch controls.

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What is User Experience (UX) Design?

User experience (UX) design is the process design teams use to create products that provide meaningful and relevant experiences to users. UX design involves the design of the entire process of acquiring and integrating the product, including aspects of branding, design, usability and function.

Designing an experience includes not only making the software easy to use but also designing the other experiences related to the product, for example, the marketing campaign, the packaging and after-sales support. Most importantly, UX design is concerned with delivering solutions that address pain points and needs. After all, no one will use a product that serves no purpose.

What UX Designers Do Goes Beyond UI Design

“User Experience Design” is often used interchangeably with terms such as “User Interface Design” and “Usability.” However, while usability and user interface (UI) design are important aspects of UX design, they are subsets of it.

A UX designer is concerned with the *entire* process of acquiring and integrating a product, including aspects of branding, design, usability and function. It’s a story that begins before the device is even in the user’s hands.

“No product is an island. A product is more than the product. It is a cohesive, integrated set of experiences. Think through all of the stages of a product or service – from initial intentions through final reflections, from the first usage to help, service, and maintenance. Make them all work together seamlessly.”

— *Don Norman, inventor of the term “User Experience.”*

Products that provide a great user experience (e.g., the iPhone) are thus designed with the product’s consumption or use in mind and the entire process of acquiring, owning and even troubleshooting it. Similarly, UX designers don’t just focus on creating usable products; they concentrate on other aspects of the user experience, such as pleasure, efficiency and fun. Consequently, there is no single definition of a good user experience. Instead, a good user experience meets a particular user’s needs in the specific context where they use the product.

A UX designer attempts to answer the question: "How can we make the experience of interacting with a computer, a smartphone, a product, or a service as intuitive, smooth and pleasant as possible?"

UX Design—A Formal Definition

The International Organization for Standardization (ISO) defines user experience as:

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“A person's perceptions and responses that result from the use or anticipated use of a product, system or service.”

— *ISO 9241-210, Ergonomics of human-system interaction—Part 210: Human-centered design for interactive systems*

We can break this definition into two parts:

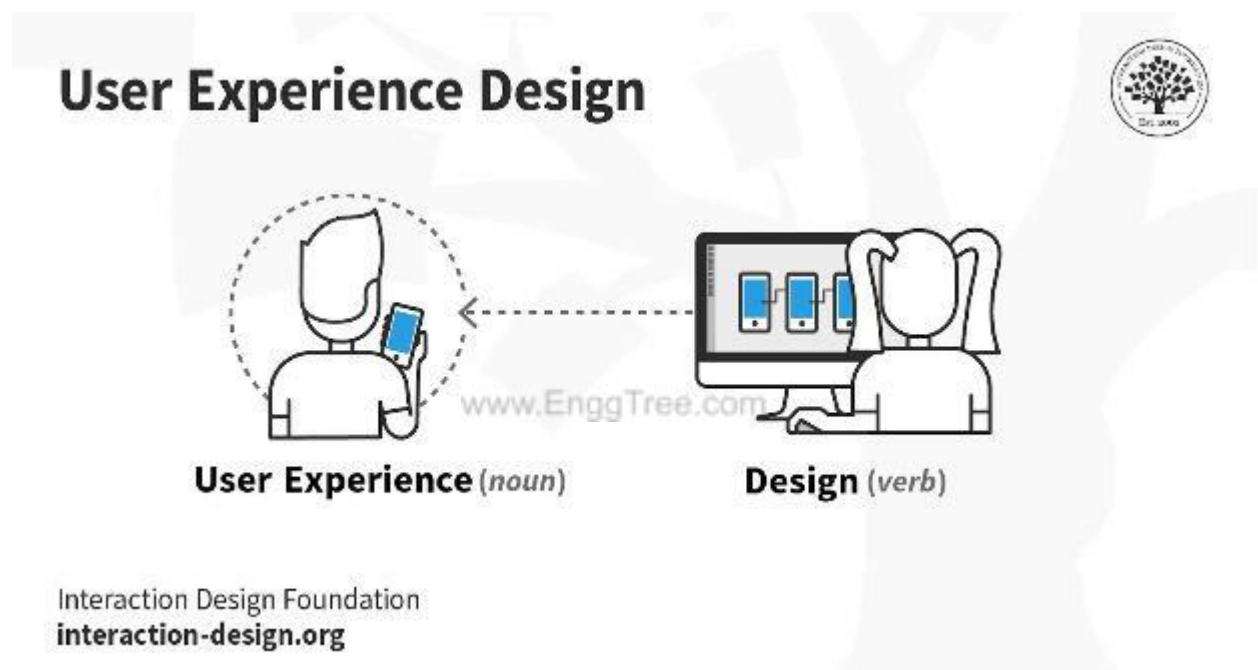
1. A person’s perceptions and responses.
2. The use of a product, system or service.

In user experience, designers do not have much control over a person’s perceptions and responses—the first part of the definition. For example, they cannot control how someone feels, moves their fingers or controls their eyes as they use a product. However, designers can control how the product, system or service behaves and looks—the second part of the definition.

“One cannot design a user experience, only design for a user experience. In particular, one cannot design a sensual experience, but only create the design features that can evoke it.”

— Jeff Johnson, Assistant Professor in the Computer Science Department of the University of San Francisco

The simplest way to think about user experience design is as a verb and a noun. A UX designer designs (verb)—ideates, plans, changes—the things that affect the user experience (noun)—perceptions and responses to a system or service.



The simplest way to think about user experience design is as a verb and a noun.

For example, when using a physical device, such as a computer mouse, we can control some aspects of the product that influence whether the user enjoys looking at, feeling and holding it:

- The way it fits in their hand. Is it snug? Is it too big and cumbersome?
- The weight. Does it affect their ability to move it as they wish?
- Its ease of use. Can they use it automatically, or do they have to think hard about it to achieve a goal?

When a person uses a digital product, such as a computer application, a few aspects that we can influence include:

- How intuitively they can navigate through the system.
- The cues that help guide them to their goal.
- The visibility of the essential aspects of a task at the appropriate time.

UX Designers Consider the Why, What, and How of Product Use

As a UX designer, you should consider the *Why*, *What* and *How* of product use. The *Why* involves the users' motivations for adopting a product, whether they relate to a task they wish to perform with it or to values and views that users associate with the ownership and use of the product. The *What* addresses the things people can do with a product—its functionality. Finally, the *How* relates to the design of functionality in an accessible and aesthetically pleasant way.

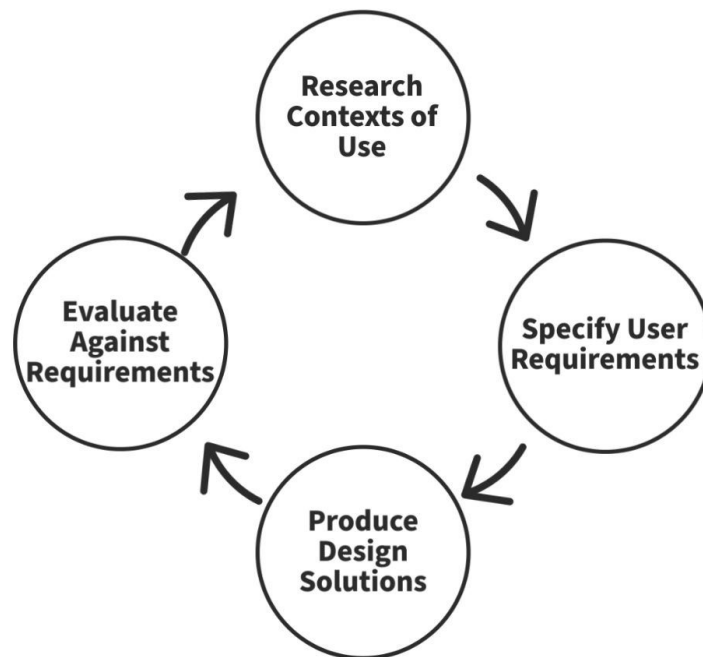
UX designers start with the *Why* before determining the *What* and then, finally, the *How* to create products that users can form meaningful experiences with. In software designs, you must ensure the product's "substance" comes through an existing device and offers a seamless, fluid experience.

UX Design is User-Centered

Since UX design encompasses the entire user journey, it's a multidisciplinary field—UX designers come from various backgrounds such as visual design, programming, psychology and interaction design. To design for human users also means working with a heightened scope regarding accessibility and accommodating many potential users' physical limitations, such as reading small text.

A UX designer's typical tasks vary but often include user research, creating personas, designing wireframes and interactive prototypes, and testing designs. These tasks can vary significantly from one organization to the next. Still, they always demand designers to be the users' advocates and keep their needs at the center of *all* design and development efforts. That's also why most UX designers work in some form of user-centered work process and keep channeling

their best-informed efforts until they address all of the relevant issues and user needs optimally.



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User-centered design is an iterative process where you take an understanding of the users and their context as a starting point for all design and development.

UI and UX

The UI is often talked about in conjunction with user experience (UX), which may include the aesthetic appearance of the device, response time and the content that is presented to the user within the context of the user interface. Both terms fall under the concept of human-computer interaction (HCI), which is the field of study focusing on the creation of computer technology and the interaction between humans and all forms of IT design. Specifically, HCI studies areas such as UCD, UI design and UX design. An increasing focus on creating an optimized user experience has led some to carve out careers as UI and UX experts. Certain languages, such as HTML and CSS, have been geared toward making it easier to create a strong user interface and experience.

Difference between UI and UX

Developing a product that people love often requires both good UI and good UX. For example, you could have a messaging app (such as WhatsApp) that looks great and has intuitive navigation (UI). But if the app loads slowly or makes you click through numerous screens to message someone (UX), it doesn't matter how good it looks. You're probably not going to want to use it.

On the other hand, a website could be loaded with unique, helpful content organised in a logical and intuitive way. But if it looks dated or you can't easily figure out how to move between screens or scroll through options, you're likely to click away from the site.



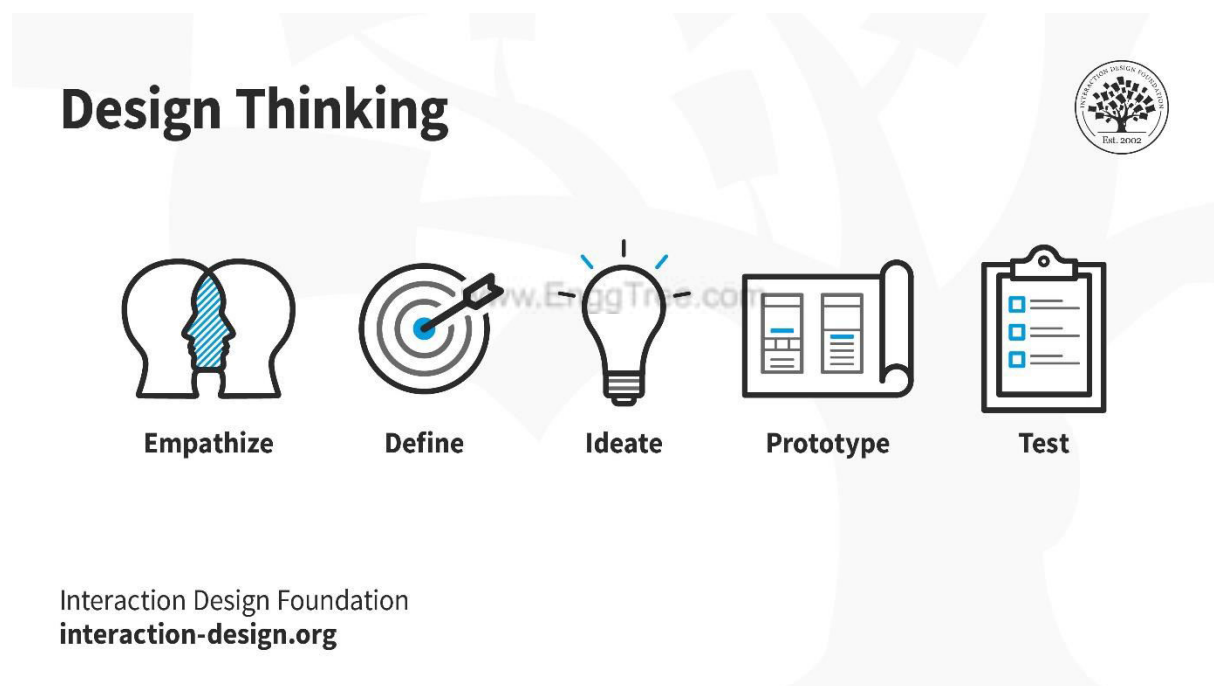
UX vs. UI designers

UX designer	UI designer
 Interaction designer	 Visual designer
 Charts the user pathway	 Chooses color and typography
 Plans information architecture	 Plans visual aesthetic
 Expert in wireframes, prototypes, and research	 Expert in mockups, graphics, and layouts

Aspect	User Interface (UI)	User Experience (UX)
Definition	UI focuses on the visual elements users interact with.	UX focuses on the overall experience users have.
Scope	Deals with the layout, design, and presentation.	Encompasses the entire journey of the user.
Purpose	Aims to provide an attractive and usable interface.	Aims to ensure the user's needs are met effectively.
Elements	Involves buttons, icons, menus, and other components.	Involves emotions, perceptions, and interactions.
Goal	Enhances the aesthetic appeal and usability of a product.	Enhances user satisfaction and fulfillment.
Interaction	Deals with how users interact with the interface.	Deals with how users feel during their interactions.
Measurability	Can be measured by visual appeal, consistency, etc.	Measured by user feedback, metrics, and satisfaction.
Medium	Primarily concerned with the visual aspects.	Involves both visual and psychological aspects.
Output	The end result is a visually appealing design.	The end result is a satisfying user experience.

Collaboration	Collaborates with UX to implement design elements.	Collaborates with UI to deliver a cohesive experience.
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Design thinking's value as a world-improving, driving force in business (global heavyweights such as Google, Apple and Airbnb have wielded it to notable effect) matches its status as a popular subject at leading international universities. **With design thinking, teams have the freedom to generate ground-breaking solutions.** Using it, your team can get behind hard-to-access insights and apply a collection of hands-on methods to help find innovative answers.



Design thinking is an iterative and non-linear process that contains five phases: 1. Empathize, 2. Define, 3. Ideate, 4. Prototype and 5. Test.

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1. Stage 1: Empathize—*Research Your Users' Needs*

Here, you should gain an empathetic understanding of the problem you're trying to solve, typically through user research. Empathy is crucial to a human-centered

design process such as design thinking because it allows you to set aside your own assumptions about the world and gain real insight into users and their needs.

2. Stage 2: Define—*State Your Users' Needs and Problems*

It's time to accumulate the information gathered during the Empathize stage. You then analyze your observations and synthesize them to define the core problems you and your team have identified. These definitions are called problem statements. You can create personas to help keep your efforts human-centered before proceeding to ideation.

3. Stage 3: Ideate—*Challenge Assumptions and Create Ideas*

Now, you're ready to generate ideas. The solid background of knowledge from the first two phases means you can start to "think outside the box", look for alternative ways to view the problem and identify innovative solutions to the problem statement you've created. Brainstorming is particularly useful here..

4. Stage 4: Prototype—*Start to Create Solutions*

This is an experimental phase. The aim is to identify the best possible solution for each problem found. Your team should produce some inexpensive, scaled-down versions of the product (or specific features found within the product) to investigate the ideas you've generated. This could involve simply paper prototyping.

5. Stage 5: Test—*Try Your Solutions Out*

Evaluators rigorously test the prototypes. Although this is the final phase, design thinking is iterative: **Teams often use the results to *redefine* one or more further problems.** So, you can return to previous stages to make further iterations, alterations and refinements – to find or rule out alternative solutions.

Overall, you should understand that **these stages are different *modes* which contribute to the entire design project, rather than sequential steps.** Your goal throughout is to gain the deepest understanding of the users and what their ideal solution/product would be.

Why Is Design Thinking so Important?

Design thinking is essential in UX design to adapt to rapid changes in users' behaviors and environments. Since its inception by Herbert A. Simon in 1969, it has evolved and been embraced across various fields. Today, organizations use it to solve complex problems and cater to human needs. Design teams leverage design thinking to tackle challenging issues and prioritize user-centric solutions. It encourages "thinking outside the box," leading to improved UX research, prototyping, and usability testing to meet users' evolving needs effectively.

Divergent Vs Convergent Thinking: What's the Difference?

Divergence and convergence is a problem-solving method that emphasizes empathy, experimentation, and iteration. It is a human-centered design approach that focuses on understanding people's needs, behaviors, and motivations to design creative solutions that meet those needs effectively.

This problem-solving process cycles between expansive and focused thinking to better understand problems people face and develop a solution. Coined by psychologist J.P. Guilford in the 1950s, these cycles of thinking provide the structure for this iterative design thinking process.

In this article we'll define divergence and convergence, provide examples, walk through each stage, suggest free templates you can use to try it out, and provide tips to help you get the most out of the creative process.

Divergence vs convergence

Divergence and convergence are like the inhale and exhale of the design thinking process. Divergence creates space and possibilities, while convergence brings focus and direction. Designers move back and forth between these two processes throughout the design thinking process, building on the insights and ideas generated during each cycle to refine and improve the solution.

Note: Divergent and convergent phases are non-linear, meaning that when facilitating this process, you may move back and forth as you discover new insights during convergence.

What is divergence?

Divergence is the process of exploring and generating a wide range of ideas and possibilities without any judgment or criticism. Divergent thinking allows designers to expand the possibilities and consider different perspectives, angles, and dimensions of the problem.

What is convergence?

Convergent thinking refers to the process of selecting, evaluating, and refining the most promising ideas from the divergent phase. It involves narrowing down and refining possibilities by selecting the best ideas and combining them to create a coherent and effective solution.

Benefits of divergent and convergent thinking

- Increased creativity and innovation
- Better decision-making
- Human-centered design
- Enhanced problem-solving
- Improved collaboration

What is Brainstorming?

- Brainstorming is a method design teams use to generate ideas to solve clearly defined design problems. In controlled conditions and a free-thinking environment, teams approach a problem by such means as “How Might We” questions. They produce a vast array of ideas and draw links between them to find potential solutions.
- Resume
- Watch how to get serious, and fun, results with Brainstorming.
- **How To Use Brainstorming Best**
- Brainstorming is part of design thinking. You use it in the ideation phase. It’s extremely popular for design teams because they can expand in all directions. Although teams have rules and a facilitator to keep them on track, they are free to use out-of-the-box and lateral thinking to seek the most effective solutions to any design problem. By brainstorming, they can take a vast number of approaches—the more, the better—instead of just exploring conventional means and running into the associated obstacles. When teams work in a judgment-free atmosphere to find the real dimensions of a problem, they’re more likely to produce rough answers which they’ll refine into possible solutions later. Marketing CEO Alex Osborn, brainstorming’s “inventor”, captured the refined elements of creative problem-solving in his 1953 book, *Applied Imagination*. In brainstorming, we aim squarely at a design problem and produce an arsenal of potential solutions. By not only harvesting our own ideas but also considering and building on colleagues’, we cover the problem from every angle imaginable.
- “It is easier to tone down a wild idea than to think up a new one.”

— Alex Osborn

- Everyone in a design team should have a clear definition of the target problem. They typically gather for a brainstorming session in a room with a large board/wall for pictures/Post-Its. A good mix of participants will expand the experience pool and therefore broaden the idea space.



- Brainstorming may seem to lack constraints, but everyone must observe eight house rules and have someone acting as facilitator.
- **Set a time limit** – Depending on the problem’s complexity, 15–60 minutes is normal.
- **Begin with a target problem/brief** – Members should approach this sharply defined question, plan or goal and stay on topic.
- **Refrain from judgment/criticism** – No-one should be negative (including via body language) about any idea.
- **Encourage weird and wacky ideas** – Further to the ban on killer phrases like “too expensive”, keep the floodgates open so everyone feels free to blurt out ideas (provided they’re on topic).
- **Aim for quantity** – Remember, “quantity breeds quality”. The sifting-and-sorting process comes later.
- **Build on others’ ideas** – It’s a process of association where members expand on others’ notions and reach new insights, allowing these ideas to trigger their own. Say “and”—rather than discourage with “but”—to get ideas closer to the problem.

Stay visual – Diagrams and Post-Its help bring ideas to life and help others see things in different ways.

Allow one conversation at a time – To arrive at concrete results, it’s essential to keep on track this way and show respect for everyone’s ideas.



- To capture everyone's ideas in a brainstorming session, someone must play "scribe" and mark every idea on the board. Alternatively, write down your own ideas as they come, and share these with the group. Often, design problems demand mixed tactics: brainstorming and its sibling approaches – **braindumping** (for individuals), and **brainwriting** and **brainwalking** (for group-and-individual mixes).
- **Take Care with Brainstorming**
- Brainstorming involves **harnessing synergy** – we leverage our collective thinking towards a variety of potential solutions. However, it's challenging to have boundless freedom. In groups, introverts may stay quiet while extroverts dominate. Whoever's leading the session must "police" the team to ensure a healthy, solution-focused atmosphere where even the shiest participants will speak up. A warm-up activity can cure brainstorming "constipation" – e.g., ask participants to list ways the world would be different if metal were like rubber.

What is gamestorming?

Gamestorming is a form of brainstorming that leverages the use of games.

As we all know, playing games enforces rules, effectively helping to overcome the usual brainstorming limitations. The way brainstorming avoids conflicts is by shifting group and thought dynamics.

To give you a real life example, at Collaboard we need **to coordinate employees across multiple locations**, and found that gamestorming (especially when facilitated through our app Collaboard) makes it easier to handle collaborative and creative processes.

We started to test it, mainly out of our love for innovation, and played the 3-12-3 Brainstorm game, ideated by **James Macanufo**, involving employees in three different locations: Zurich, Munich and Milano. Keep reading to discover this amazing technique.

The 3-12-3 Brainstorming: a new approach to creative collaboration
So, how does this game work? There can be up to 10 players, who must follow these rules:

- First of all, they must choose a **two-word topic** to focus on.
- After the choice, phase one of the game begins: for 3 minutes, each team member **brainstorms** on his own and writes down their ideas on sticky notes.
- When everyone's done brainstorming, the team is divided into sub-teams of 2 people each. This is the beginning of phase 2.
- For 12 minutes, teams pick three random cards and develop a **full-fledged concept**.
- In phase 3, teams have 3 minutes each to present their ideas to everyone else.

All you need is Collaboard!

So, back to our experiment: we decided to brainstorm some ideas for Christmas Cards. For the sake of simplicity, we split our team into 3 sub-teams according to their physical location: Berlin, Zurich, Munich.



Phase One

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This first phase is all about **generating ideas**. Each team member is on his own and must write on sticky notes whatever idea comes to mind.

This part felt very similar to classic brainstorming, but we quickly noticed two differences:

- Each person was using their own device, so they didn't fear judgment from their peers, and didn't forget what they wanted to say/write while waiting for their turn.
- The limited time of 3 minutes meant that there was no space for self-doubt. As sticky notes were being written, they were quickly moved into a grid library.



Phase Two

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Once solo brainstorming was completed, each team member selected three random stick notes from the library to start developing concepts around them.

We found this phase a bit more challenging than the first one, as the cards could involve clashing ideas. But this is exactly where collaboration steps in: the team has to work together in order to consolidate the ideas into a final, collaborative project.

With **Collaboard by our side**, visualizing and working through sketches, images, prototypes and concepts made the process way easier

Phase Three

Finally, we approached the final phase. Using sketches created in phase two, we proceeded to present our final ideas to the team.

The **integration with Skype for Business** made the remote presentation extremely easy - we ended up with three different proposals (ready to be approved by the marketing manager) in less than 30 minutes, which is hard to imagine with a traditional brainstorming session!

Also, we all could leverage other teams' ideas, nurturing collaboration and strengthening our connection.

Observational empathy is a crucial aspect of the design thinking process that involves closely observing and understanding the experiences, behaviors, and needs of users by immersing oneself in their environment. It is a method of gaining deep insights into users' lives, thoughts, and emotions, which helps designers create more meaningful and user-centered solutions.

In observational empathy, designers go beyond simply asking users about their preferences or needs. Instead, they spend time directly observing users in their natural context, without interrupting or interfering with their actions. This approach allows designers to understand how users interact with products, services, or environments and how they overcome challenges or frustrations.

Observational empathy

Key aspects of observational empathy include:

1. ****Active Observation****: Designers keenly observe users' behaviors, gestures, facial expressions, and body language to understand their emotions and reactions.
2. ****Listening and Note-taking****: They listen attentively to users' conversations and take notes on their experiences, pain points, and unmet needs.

3. ****Contextual Understanding****: Observational empathy takes place in the real environment where users use the product or service, providing context-rich information for design decisions.

4. ****Identifying Unspoken Needs****: Through observation, designers can identify needs that users may not explicitly communicate or may not be aware of themselves.

5. ****Avoiding Assumptions****: Observational empathy helps designers avoid making assumptions about users' preferences or behaviors, as it relies on direct evidence.

By practicing observational empathy, designers can gain a deeper and more holistic understanding of their target audience. This empathetic understanding forms the foundation for creating solutions that truly resonate with users and provide a positive impact on their lives. It is an essential step in the design thinking process, enabling designers to design with empathy and solve real problems effectively.

Unit-2

Foundations of UI Design

Visual and UI Principles- UI Elements and Patterns – Interaction Behaviors and Principles –Branding- Style Guides

Visual Design Principles

When we look at something, we often instinctively know if it looks good or not, thanks to the emotional response it elicits. Though we may struggle to put it into words, certain design elements can make visuals more appealing and effective in user experience.

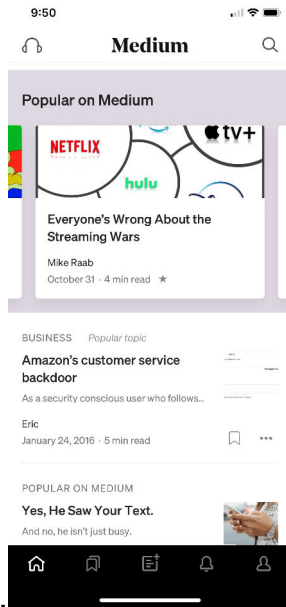
Here are 5 key visual design principles that impact UX:

1. **Scale:** The relative size of elements in a design. It helps create emphasis, guide attention, and add visual interest.
2. **Visual Hierarchy:** The arrangement of elements to establish a clear order of importance. It guides users through content and makes information easier to understand.
3. **Balance:** Distributing visual weight evenly in a design to achieve stability. It ensures that no single element overpowers others, leading to a harmonious composition.
4. **Contrast:** Differences in color, size, or other attributes that create distinction between elements. Contrast enhances readability and draws attention to essential information.
5. **Gestalt:** The idea that our brains perceive elements as a whole rather than individual parts. It emphasizes how elements are organized to form a unified and coherent design.

By adhering to these principles, designers can create engaging and user-friendly visuals that enhance the overall experience.

1. Scale

Scale in visual design involves using relative size to communicate importance and rank within a composition. Important elements are made larger than less significant ones, making them more noticeable. A visually appealing design typically uses up to three different sizes to create variety and establish a visual hierarchy. By emphasizing essential elements through size, users can easily understand and interact with the design

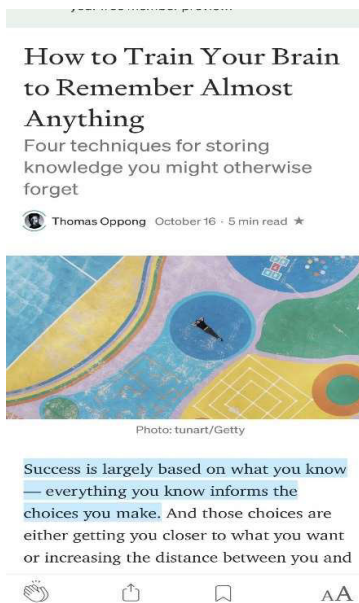


Medium for iPhone: Popular articles are visually larger than other articles. The scale directs users to potentially more-interesting article.

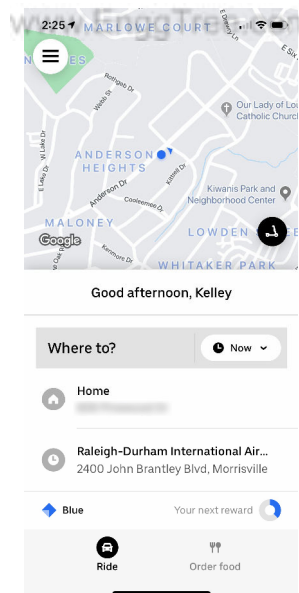
2. Visual Hierarchy

Visual hierarchy is about guiding the user's eye to focus on important elements first. It's achieved by using various design cues like size, color, spacing, and placement. A good layout with clear visual hierarchy makes it easy for users to understand the content and its importance. You can use different font sizes, bright colors for important items, and muted colors for less important ones to create a clear hierarchy.

Including small, medium, and large elements in the design helps establish a balanced scale and reinforce the visual hierarchy..



Medium mobile app: There is a clear visual hierarchy of title, subtitle, and body text. Each component of the article is in a type size equal to its importance.



Uber mobile app: The visual hierarchy is clear in Uber's mobile app. The screen is split in half between the map and input form (bottom half of screen), which enforces the thought that these components are equally important to the user. The eye is immediately drawn to the Where to? field because of its gray background, then to the recent locations below it, which are slightly smaller in font size

3. Balance

Balance in visual design is like a seesaw, where you arrange elements to create a pleasing proportion. It means having an equal distribution of visual elements on both sides of an imaginary line running through the middle of the screen. This line can be vertical or horizontal.

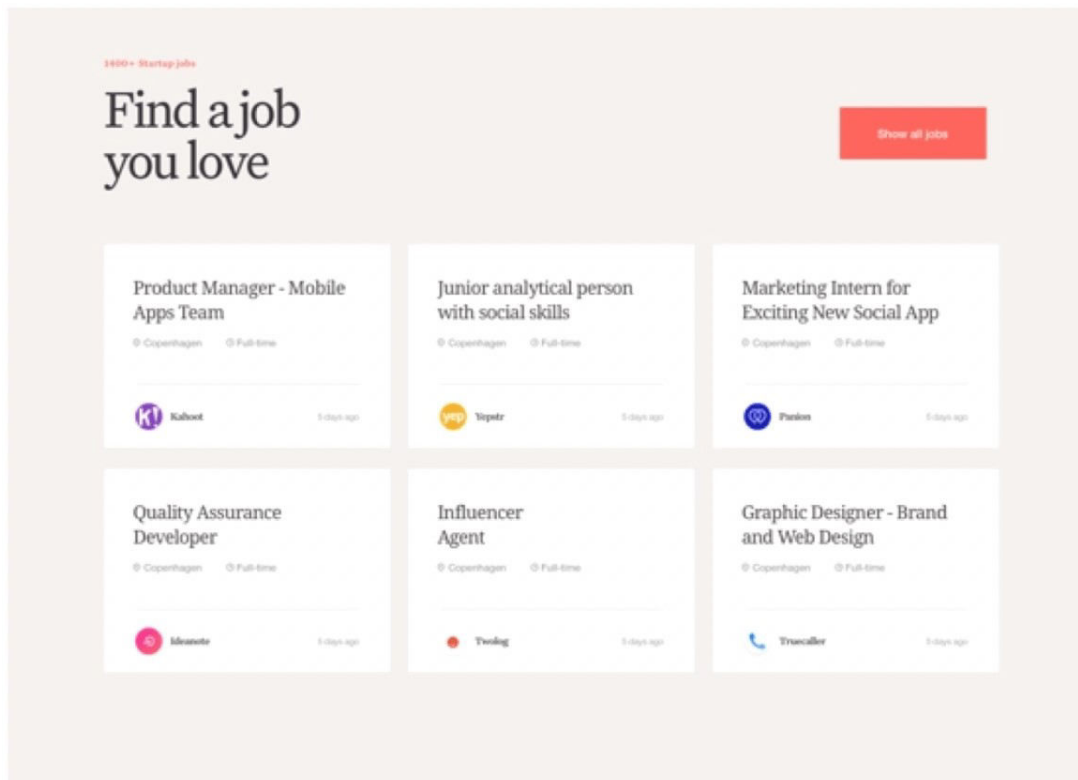
To achieve balance, you need to consider the size and space taken by elements, not just their number. If one side has a large element and the other side has a small one, the design feels unbalanced.

There are three types of balance:

1. Symmetrical - elements are evenly distributed on both sides of the central axis.
2. Asymmetrical - elements are unevenly distributed, creating a sense of movement and energy.
3. Radial - elements radiate out from a central point, guiding the eye to the center of the composition.

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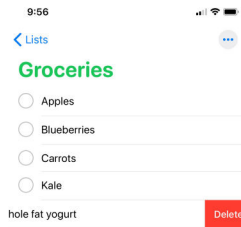
The choice of balance depends on the message you want to convey. Asymmetry is dynamic and engaging, while symmetry is calm and static. Radial balance always draws attention to the center. Achieving balance ensures that no area overwhelms others, creating a harmonious and pleasing design.



The Hub Style Exploration: The composition feels stable, which is especially appropriate when you're looking for a job you love. The balance here is symmetrical. If you were to draw an imaginary vertical axis down the center of the website, elements are distributed equally on both sides of the axis. (Image source: dribbble.com)

4. Contrast

Contrast is a powerful principle in design that highlights differences between elements. It involves placing visually dissimilar things together to show that they belong in different categories or serve distinct purposes. For example, using bright colors like red to signify deletion in a UI design makes the element stand out from the rest, emphasizing its unique function. Contrast helps users quickly identify and understand important information, making the design more effective and user-friendly



+ New Reminder

Reminders app on iOS: The color red, which has high contrast to its surrounding context, is reserved for deleting.

Often, in UX the word “contrast” brings to mind the contrast between text and its background. Sometimes designers deliberately decrease the text contrast in order to deemphasize less important text. But this approach is dangerous — reducing text contrast also reduces legibility and may make your content inaccessible. Use a color-contrast checker to ensure that your content can still be read by all your target users.



Greenhouse Juice Co: The legibility of the text on the bottle relies on the color of juice. Although the contrast works beautifully for some juices,

labels for bottles with light colored juices are nearly impossible to read.

(Image source: www.instagram.com)

5. Gestalt Principles

Gestalt principles are a set of ideas from psychology that explain how humans make sense of complex images. Instead of seeing individual parts, our minds automatically organize them into a whole. This means we tend to perceive the complete picture rather than separate elements.

These principles include similarity, continuation, closure, proximity, common region, figure/ground, and symmetry and order. One important principle for UX is proximity, which means that items visually close together are seen as belonging to the same group. These principles help designers create visuals that are easy for users to



understand and navigate.

It is the Gestalt closure principle that allows us to see two figures kissing instead of random shapes in Picasso's painting. Our brains fill in the missing pieces to create two figures.



We also often see applications of the Gestalt theory in logos. In the NBC logo, there is no peacock in the white space, but our brain understands there to be one.

UI Design Principle:

The seven principles of universal design are: Equitable use, flexibility in use, simple and intuitive use, perceptible information, tolerance for error, low physical effort, and size and space for approach and use.

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Equitable Use: The design should be usable by all individuals, regardless of their age, abilities, or background. It ensures that everyone can access and interact with the product or environment in a fair and inclusive manner.

2. **Flexibility in Use:** The design should accommodate a wide range of individual preferences and abilities. It offers multiple ways to interact with the product or environment, allowing users to choose the approach that suits them best.
3. **Simple and Intuitive Use:** The design should be easy to understand and use, requiring minimal effort to figure out its operation. Clear instructions and logical layouts make it intuitive for users to navigate and perform tasks.
4. **Perceptible Information:** The design should present information effectively to all users, regardless of their sensory abilities. It ensures that important information is available through multiple senses (e.g., visual, auditory, tactile) so that users can perceive and comprehend it easily.
5. **Tolerance for Error:** The design should minimize the risk of errors and provide ways to recover from mistakes. It avoids critical consequences for unintended actions and offers clear feedback to guide users in correcting errors.
6. **Low Physical Effort:** The design should require minimal physical effort to use. It considers the needs of users with varying physical abilities, making interactions comfortable and not overly demanding on strength or dexterity.
7. **Size and Space for Approach and Use:** The design should provide adequate space and reach ranges to accommodate users of different sizes and mobility. It ensures that individuals using mobility aids or who have limited reach can comfortably access and use the design.

UI Elements

UI elements are the building blocks of a user interface, the individual parts that users see and interact with on websites, apps, and other digital product

1. Buttons: Buttons are like interactive squares or circles that you can click or tap on. They perform actions like "Submit," "Next," or "Cancel."
2. Text Fields: Text fields are areas where you can type in text, like when you search for something or enter your name.
3. Checkboxes: Checkboxes are like little boxes that you can tick or un-tick to make a choice from a list of options.
4. Radio Buttons: Radio buttons are like small circles or dots that you can select, but you can only choose one option from a group of choices.
5. Icons: Icons are small pictures or symbols that represent actions or ideas, like a magnifying glass for search or a home symbol for the homepage.
6. Dropdown Menus: Dropdown menus are like hidden lists that appear when you click on a button or a label. They show you more options to choose from.
7. Sliders: Sliders are like bars or scales that you can slide left or right to adjust settings or choose a value.
8. Tabs: Tabs are like labeled sections that you can click on to switch between different parts of a website or app.
9. Tooltips: Tooltips are like small pop-up boxes that show extra information when you hover your mouse over something.
10. Cards: Cards are like little containers that hold information, like a picture, a title, and a short description, neatly displayed together.

These UI elements help users interact with websites and apps by clicking, typing, or selecting options, making the overall experience more intuitive and enjoyable.

UI Patterns

User interface (UI) design patterns are reusable/recurring components which designers use to solve common problems in user interface design. For example, the breadcrumbs design pattern lets users retrace their steps. Designers can apply them to a broad range of cases, but must adapt each to the specific context of use.

Common UI Design Patterns

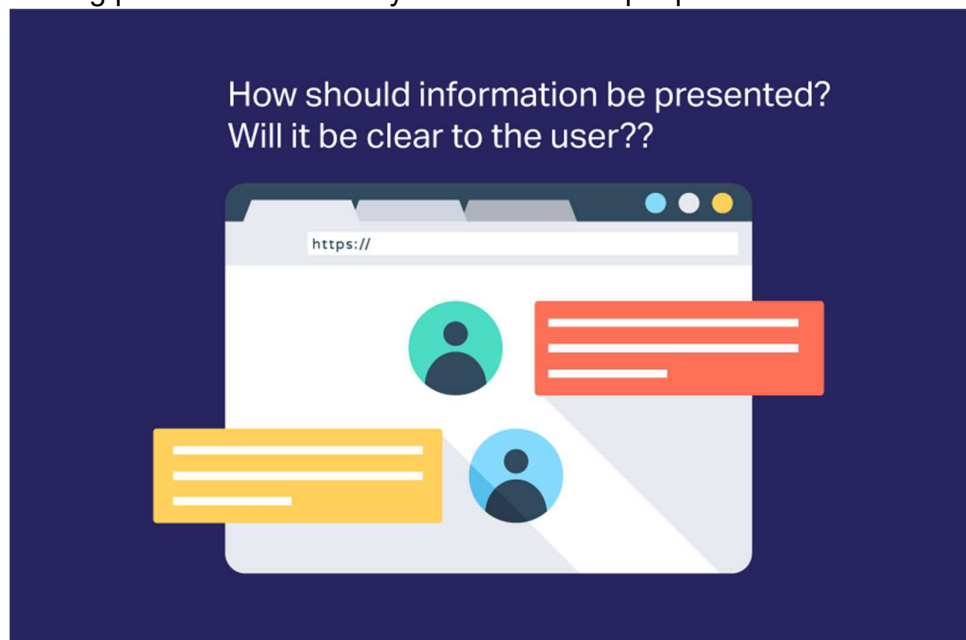
Some of the most common UI design patterns are:

- **Breadcrumbs** – Use linked labels to provide secondary navigation that shows the path from the front to the current site page in the hierarchy.
- **Lazy Registration** – Forms can put users off registration. So, use this sign-up pattern to let users sample what your site/app offers for free or familiarize themselves with it. Then, you show them a sign-up form. For example, Amazon allows unrestricted navigation and cart-loading *before* it prompts users to register for an account. Note:
 - When content is accessible only to registered users or users must keep entering details, offer them **simplified/low-effort sign-up forms**.
 - **Minimize/Avoid optional information fields**. Use the **Required Field Markers** pattern to guide users to enter needed data.
- **Forgiving Format** – Let users enter data in various formats (e.g., city/town/village or zip code).
- **Clear Primary Actions** – Make buttons stand out with color so users know what to do (e.g., “Submit”). You may have to decide which actions take priority.
- **Progressive Disclosure** – Show users only features relevant for the task at hand, one per screen. If you break input demands into sections, you’ll reduce cognitive load (e.g., “Show More”).
- **Hover Controls** – Hide nonessential information on detailed pages to let users find relevant information more easily.
- **Steps Left** – Designers typically combine this with a wizard pattern. It shows how many steps a user has to take to complete a task. You can use gamification (an incentivizing design pattern) here to enhance engagement.
- **Subscription Plans** – Offer users an options menu (including “Sign-up” buttons) for joining at certain rates.
- **Leaderboard** – You can boost engagement if you use this social media pattern.

- **Dark Patterns** – Some designers use these to lead or trick users into performing certain actions, typically in e-commerce so they spend more or surrender personal information. Dark patterns range in harmfulness. Some designers leave an unchecked opt-out box as a default to secure customer information. Others slip items into shopping carts. To use dark patterns responsibly, you must be ethical and have empathy with your users. Dark patterns are risky because user mistrust and feedback can destroy a brand's reputation overnight.

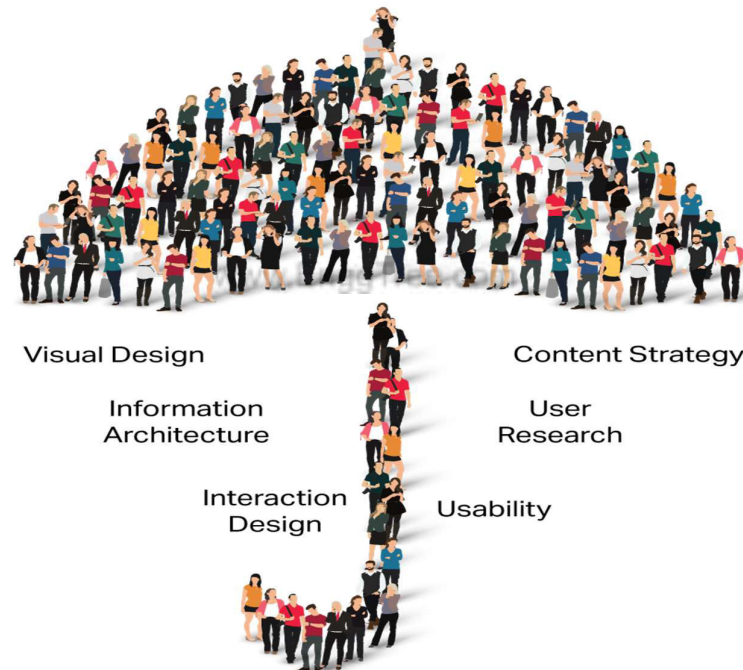
What is Interaction Design?

- Interaction design (IxD) is about designing how users interact with products, especially software products like apps and websites. The main goal is to create products that help users achieve what they want in the easiest and most effective way possible.
- Interaction designers work on various elements like how things look, move, sound, and the space they take up. For example, they might create sounds for user actions. They think about whether the product behaves in a way that users expect and if users can achieve their goals without much effort.
- In some cases, UX designers are also responsible for interaction design along with other aspects of product design. Overall, interaction design is about making products user-friendly and intuitive for people to use.



How Does Interaction Design Differ from UX Design?

- Interaction design and UX design have similarities, but they are not the same thing. Interaction design focuses on how users interact with a product, making sure the interactions are smooth and easy. It deals with elements like how things move, look, and sound to create a good user experience.
- On the other hand, UX design is broader. It involves understanding the users through research, creating user personas to know their needs, and testing the product with real users. UX designers ensure the entire experience of using the product is great, not just the interactions.
- In simple terms, interaction design is a part of UX design, but UX design covers more aspects beyond just the interactions.



- *Interaction design is a part of UX design, which is about creating a great experience for users when they use a product. Interaction designers focus on improving the way users interact with the product. For example, making sure the TV remote or doorknob is easy to use.*
- *UX designers look at the whole user journey, not just the moment of interaction. They consider all aspects of the product that users interact with. Interaction design is like a small part of the bigger picture of user experience design. It's impossible to have good interaction design without considering the overall user experience.*

Why is Interaction Design Important?

In today's digital world, creating a great interaction experience is crucial. It can be as simple as using a TV remote or as complex as using software. Smart tech companies understand that a pleasant interaction leaves users satisfied or delighted. Remember the first time you used an iPhone?

Steve Jobs, the founder of Apple, emphasized that design is not just about looks but also about how things work and how users feel about them.

Here are some important interaction design principles:

1. Match user experience and expectations.
2. Maintain consistency in design throughout the application.
3. Follow functional minimalism by keeping things simple and useful.
4. Reduce mental pressure and make the application easy to understand.
5. Design interactively to keep users engaged.
6. Allow users to have control, trust, and explore the product.
7. Use intuitive and interactive media to invite interactions.
8. Make user interactions easy to learn and remember.
9. Take care to prevent errors and handle them effectively if they occur.
10. Use familiar and physical-world interactions to make actions feel natural.

The interaction design process involves five stages:

1. Discover users' real needs by observing, interviewing, and analyzing existing solutions.
2. Design a potential solution based on design guidelines and principles, using techniques that match users' interaction preferences.

3. Start prototyping to give users a glimpse of the product and gather feedback for improvement.
4. Implement and deploy the final version of the product.
5. The process is iterative, meaning it may take multiple rounds of testing and refinement to reach the best solution.

Understanding your users' needs and continuously adapting to their feedback is vital in the interaction design process. Design thinking is a helpful approach to gain insights and create features that truly serve users' needs and desires. By empathizing with users, designers can identify their real-world needs and provide better solutions.

Branding

Branding in UI design is the process of incorporating a company's visual identity and personality into the user interface of a product or website. It involves using consistent colors, typography, logos, and other design elements to create a recognizable and cohesive brand image. Branding is essential in UI design as it helps users associate the product with the company and builds trust and familiarity.

****The Need for Branding in UI Design:****

1. ****Recognition:**** Consistent branding elements help users identify the product with the company, even across different platforms and touchpoints.
2. ****Trust and Credibility:**** A strong brand presence instills confidence in users, making them more likely to engage with the product.
3. ****Consistency:**** Branding ensures a unified and cohesive experience, making the product feel more reliable and professional.
4. ****Differentiation:**** In a competitive market, branding sets the product apart from competitors, making it stand out in users' minds.

****The Stages of Branding in UI Design:****

1. ****Brand Strategy:**** Define the brand's core values, target audience, and unique selling proposition (USP). This stage sets the foundation for the visual elements that will be used in the UI design.
2. ****Visual Identity Design:**** Create the visual elements that represent the brand, such as the logo, color palette, typography, and iconography. These elements should align with the brand strategy.
3. ****UI Design Integration:**** Incorporate the brand's visual identity into the UI design. Use the chosen colors, typography, and other elements to create a consistent and on-brand interface.
4. ****Brand Guidelines:**** Document the brand guidelines that outline how the visual elements should be used in the UI design. These guidelines ensure consistency across different designers and projects.
5. ****User Testing:**** Test the UI design with real users to ensure that the branding elements resonate with the target audience and provide a positive user experience.

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By integrating branding into the UI design process, companies can create a memorable and cohesive user experience that reflects their identity and values, ultimately leading to increased user engagement and loyalty.

Style Guide

Style guides in UI design are like detailed manuals that explain how a product or website should look and feel. They help designers, developers, and others follow consistent design rules for colors, fonts, icons, and more.

The need for style guides includes:

1. Making sure everything in the design looks the same and works well.

2. Saving time and effort by giving a reference for designers and developers to follow.
3. Keeping a consistent brand image by defining how logos and colors should be used.

The stages of creating style guides are:

1. Doing research about the brand and target audience.
2. Creating the visual elements like colors and icons for the design.
3. Making a library of reusable components like buttons and cards.
4. Writing down all the design rules and guidelines in a document.
5. Testing the design with real users and making improvements.
6. Integrating the style guide into the design and development process.

By using style guides, UI designers can create a user-friendly and cohesive interface that represents the brand well and gives a positive user experience.

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